

Notes on Axionic wormholes

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1 Action principle and equations of motion

Our starting point is the following action

$$\mathcal{S} = \int d^D x \sqrt{g} \left[\frac{1}{2\kappa} (-\mathcal{R} + 2\Lambda) + \frac{1}{2(D-1)!} H_{\mu\dots\nu} H^{\mu\dots\nu} \right]. \quad (1.1)$$

H is a $(D-1)$ -form field strength satisfying the Bianchi identity $dH = 0$. Thus, one can always write locally $H = dB$ where B is a $(D-2)$ -form field. The equations of motion are simply $d\star H = 0$. We can see classically that the dynamics of H is locally governed by a pseudo-scalar field χ defined as $\star H = i d\chi$.

That holds at the quantum level. To see this we introduce χ as a Lagrangian multiplier in the following 1st class formalism action

$$\begin{aligned} \mathcal{S}_{1st} &= \int d^D x \frac{\sqrt{g}}{2\kappa} (-\mathcal{R} + 2\Lambda) + \frac{1}{2} \int H \wedge \star H + \int \chi \wedge dH \\ &= \int d^D x \frac{\sqrt{g}}{2\kappa} (-\mathcal{R} + 2\Lambda) + \frac{1}{2} \int H \wedge \star H - \int d\chi \wedge H + \mathcal{S}_{\partial M}. \end{aligned} \quad (1.2)$$

where

$$\mathcal{S}_{\partial M} = \int_{\partial M} \chi \wedge H. \quad (1.3)$$

The variational principle is well defined if we impose Dirichlet boundary conditions for χ and for H . The path-integral whose action is defined by (1.1) can be recovered by integrating out χ imposing Bianchi's identity. In the meantime, the action is quadratic in H so we can integrate out H as well by solving its equation of motion

$$H = \star d\chi \Rightarrow \star H = (-1)^{D-1} d\chi. \quad (1.4)$$

Indeed, inserting back into the action we get

$$\mathcal{S}_\chi = \int d^D x \frac{\sqrt{g}}{2\kappa} (-\mathcal{R} + 2\Lambda) - \frac{1}{2} \int d\chi \wedge \star d\chi - \int_{\partial M} \chi \wedge \star d\chi. \quad (1.5)$$

Instanton configurations have the effect of modifying the action by mean of a non-perturbative potential $V_{np}(\chi)$ so that the total action in the pseudo-scalar variable reads

$$\mathcal{S} = \int d^D x \sqrt{g} \left[\frac{1}{2\kappa} (-\mathcal{R} + 2\Lambda) - \frac{1}{2} \partial_\mu \chi \partial^\mu \chi + V_{np}(\chi) \right] + \mathcal{S}_{GHY} + \mathcal{S}_{\partial M}. \quad (1.6)$$

We look for homogeneous and isotropic solutions of the form

$$ds^2 = N^2(\tau) d\tau^2 + a^2(\tau) d\Sigma_d^2, \quad \chi = \chi(\tau). \quad (1.7)$$

We obtain the following expressions for the determinant of the metric and the Ricci scalar

$$\sqrt{g} = N a^d, \quad \mathcal{R} = -\frac{2d}{N} \frac{d}{d\tau} \left(\frac{a'}{aN} \right) + d(d-1) \frac{k}{a^2} - d(d+1) \left(\frac{a'}{aN} \right)^2. \quad (1.8)$$

Finally, let's also mention the form of the Gibbons-Hawking term

$$\mathcal{S}_{GHY} = -\frac{1}{\kappa} \int_{\partial M} d^d x \sqrt{h} K = -\frac{1}{\kappa} \frac{\partial_\tau}{N} V_d(a^d) = -\frac{dV_d a' a^{d-1}}{\kappa N}. \quad (1.9)$$

Hence the Lagrangian density of the Einstein-Hilbert part of the action reads

$$\begin{aligned} \mathcal{L}_{EH} &= -\frac{Na^d}{2\kappa} \left[-\frac{2d}{N} \frac{d}{d\tau} \left(\frac{a'}{aN} \right) + d(d-1) \frac{k}{a^2} - d(d+1) \left(\frac{a'}{aN} \right)^2 + 2\Lambda \right] \\ &= -\frac{1}{2\kappa} \left[\frac{d(d-1)(a')^2 a^{d-2}}{N} + d(d-1) N k a^{d-2} - 2N \Lambda a^d \right] + \frac{d}{d\tau} \left[\frac{da^{d-1} a'}{\kappa N} \right]. \end{aligned} \quad (1.10)$$

The boundary term coming from this total derivative will get exactly canceled by the Gibbons-Hawking term (see (1.9)). We also have

$$\mathcal{L}_\chi = -\frac{a^d}{2N} (\partial_\tau \chi)^2 + Na^d V_{np}(\chi). \quad (1.11)$$

Also, note that taking the trace of Einstein equation allows one to evaluate the on-shell action as a function of the on-shell solution χ . Indeed,

$$\begin{aligned} G = \kappa T^\chi - \Lambda D &\Rightarrow \mathcal{R} \left(1 - \frac{D}{2} \right) = \kappa T^\chi - \Lambda D, \\ T_{\mu\nu}^\chi &= -\partial_\mu \chi \partial_\nu \chi - g_{\mu\nu} \left(-\frac{1}{2} (\partial_\lambda \chi)^2 + V_{np}(\chi) \right) \Rightarrow T^\chi = \left(\frac{D}{2} - 1 \right) (\partial_\lambda \chi)^2 - D V_{np}(\chi). \end{aligned} \quad (1.12)$$

After some manipulations one obtains

$$\frac{-\mathcal{R} + 2\Lambda}{2\kappa} = \frac{1}{2} (\partial_\lambda \chi)^2 - \frac{D V_{np}}{D-2} - \frac{2\Lambda}{\kappa(D-2)}. \quad (1.13)$$

We also have the additional boundary terms

$$\mathcal{S}_{GHY} = -\frac{dV_d a' a^{d-1}}{\kappa N} \Big|_{\partial M}, \quad \mathcal{S}_{\partial M} = -\frac{V_d}{N} a^d \chi \partial_\tau \chi \Big|_{\partial M}. \quad (1.14)$$

The on-shell action thus takes the simple expression

$$S^{\text{on-shell}} = V_d \int d\tau N a^d \left(-\frac{2V_{np}}{D-2} - \frac{2\Lambda}{\kappa(D-2)} \right) + \mathcal{S}_{GHY} + \mathcal{S}_{\partial M}. \quad (1.15)$$

This on-shell action is an important quantity since the final question one should answer is how it evolves as a function of the number of bubbles. The flow parameter is $u = N\tau$. Remark that the on-shell action does not depend on the kinetic term and can be easily deduced from this formula once χ and the scale factor is known.

Let's now derive the equation of motions. The simplest one is obtained by varying wrt to χ yielding

$$\partial_\tau \left(\frac{a^d}{N} \partial_\tau \chi \right) + Na^d \partial_\chi V_{np}(\chi) = 0. \quad (1.16)$$

Varying with respect to a in $\mathcal{L}_{EH} + \mathcal{L}_\chi$ yields

$$\begin{aligned} &\left[-\frac{da^{d-1}}{2N} (\partial_\tau \chi)^2 + dNa^{d-1} V_{np}(\chi) - \frac{1}{2\kappa} \left(\frac{d(d-1)(d-2)}{N} a'^2 a^{d-3} \right. \right. \\ &\quad \left. \left. + d(d-1)(d-2)kNa^{d-3} - 2N\Lambda da^{d-1} \right) \right] \delta a - \frac{1}{2\kappa} \frac{d(d-1)2a'a^{d-2}}{N} \delta(a'). \end{aligned} \quad (1.17)$$

Thus, integrating by parts the second term

$$\left[-\frac{da^{d-1}}{2N}(\partial_\tau\chi)^2 + dNa^{d-1}V_{\text{np}}(\chi) - \frac{1}{2\kappa}\left(\frac{d(d-1)(d-2)}{N}a'^2a^{d-3} + d(d-1)(d-2)kNa^{d-3} - 2N\Lambda da^{d-1} - 2d(d-1)\frac{d}{d\tau}\left(\frac{a'a^{d-2}}{N}\right)\right)\right]\delta a. \quad (1.18)$$

We thus get the following equation of motion

$$\frac{da^{d-1}}{2N}(\partial_\tau\chi)^2 - dNa^{d-1}V_{\text{np}}(\chi) + \frac{1}{2\kappa}\left(\frac{d(d-1)(d-2)}{N}a'^2a^{d-3} + d(d-1)(d-2)kNa^{d-3} - 2N\Lambda da^{d-1} - 2d(d-1)\frac{d}{d\tau}\left(\frac{a'a^{d-2}}{N}\right)\right) = 0. \quad (1.19)$$

Finally, varying wrt to N we generate a third (non-independent) equations

$$a^dV_{\text{np}}(\chi) + \frac{a^d}{2N^2}(\partial_\tau\chi)^2 - \frac{1}{2\kappa}\left[d(d-1)ka^{d-2} - 2\Lambda a^d - \frac{d(d-1)a'^2a^{d-2}}{N^2}\right] = 0. \quad (1.20)$$

Let's summarize the set of equations here for general (k, Λ, N)

$$\begin{aligned} \partial_\tau\left(\frac{a^d}{N}\partial_\tau\chi\right) + Na^d\partial_\chi V_{\text{np}}(\chi) &= 0, \\ \frac{d(d-1)}{a^2}\left(\frac{a'^2}{N^2} - k\right) + 2\kappa\left(\frac{(\partial_\tau\chi)^2}{2N^2} + V_{\text{np}}(\chi) + \frac{\Lambda}{\kappa}\right) &= 0, \\ \frac{(d-1)(d-2)}{a^2}\left(k + \frac{a'^2}{N^2}\right) - \frac{2(d-1)}{Na^{d-1}}\frac{d}{d\tau}\left(\frac{a'a^{d-2}}{N}\right) + 2\kappa\left(\frac{(\partial_\tau\chi)^2}{2N^2} - V_{\text{np}}(\chi) - \frac{\Lambda}{\kappa}\right) &= 0. \end{aligned} \quad (1.21)$$

In the $N = 1$ gauge, it yields

$$\begin{aligned} \partial_\tau(a^d\partial_\tau\chi) + a^d\partial_\chi V_{\text{np}}(\chi) &= 0, \\ \frac{d(d-1)}{a^2}(a'^2 - k) + 2\kappa\left(\frac{(\partial_\tau\chi)^2}{2} + V_{\text{np}}(\chi) + \frac{\Lambda}{\kappa}\right) &= 0, \\ \frac{(d-1)(d-2)}{a^2}(k + a'^2) - \frac{2(d-1)}{a^{d-1}}\frac{d}{d\tau}(a'a^{d-2}) + 2\kappa\left(\frac{(\partial_\tau\chi)^2}{2} - V_{\text{np}}(\chi) - \frac{\Lambda}{\kappa}\right) &= 0. \end{aligned} \quad (1.22)$$

Non-running solutions A trivial set of solutions can be obtained by setting χ to a extremum of the potential. These solutions will benchmark of the more general solutions. Let $V_{\text{np}}^{\text{ext}}$ an extremum of the non-perturbative potential. The geometry of the solution depends crucially on the sign of $\Lambda_{\text{eff}} = \kappa V_{\text{np}}^{\text{ext}} + \Lambda$. We work in the $N = 1$ gauge.

- If $\Lambda_{\text{eff}} > 0$

- If $k = 1 \Rightarrow a(\tau) = \frac{1}{H_{dS}}\sin(H_{dS}\tau)$, with $H_{dS} = \sqrt{\frac{2\Lambda_{\text{eff}}}{d(d-1)}} \Rightarrow EdS_{d+1}$ in spherical slices.
- Otherwise, there are no solutions for a real.

- If $\Lambda_{\text{eff}} < 0$

- If $k = 1 \Rightarrow a(\tau) = \frac{1}{H_{AdS}}\sinh(H_{AdS}\tau)$, with $H_{AdS} = \sqrt{-\frac{2\Lambda_{\text{eff}}}{d(d-1)}} \Rightarrow EAdS_{d+1}$ in spherical slices.

- If $k = -1 \Rightarrow a(\tau) = \frac{1}{H_{AdS}} \cosh(H_{AdS}\tau) \Rightarrow EAdS_{d+1}$ in hyperbolic slices.
- If $k = 0 \Rightarrow a(\tau) = \frac{1}{H_{AdS}} \exp(H_{AdS}\tau) \Rightarrow EAdS_{d+1}$ in flat slices.
- If $\Lambda_{\text{eff}} = 0$
 - If $k = 1 \Rightarrow a(\tau) = \tau \Rightarrow \mathbb{R}_{d+1}$. These are cylinder coordinates associated to Euclidean space.
 - If $k = 0 \Rightarrow a(\tau) = \text{const.}$ Upon rescaling the transverse flat slices, these are standard cartesian coordinates.

We now focus on compact=spherical slices to obtain a finite action. We could also regulate the infinite volume of the transverse slices but we postpone this issue for latter. In these non-running solutions, the boundary term from the axion part trivially vanishes. We also observe that the bulk on-shell action always take the simple form

$$S_{\text{bulk}}^{\text{on-shell}} = -\frac{2V_d\Lambda_{\text{eff}}}{\kappa(D-2)} \int d\tau a^d. \quad (1.23)$$

- If $\Lambda_{\text{eff}} > 0$: There is no boundary since the geometry is a sphere and thus the Gibbons-Hawking York term trivially vanishes. One finds

$$\begin{aligned} S^{\text{on-shell}} &= -\frac{V_d\Lambda_{\text{eff}}}{\kappa(D-2)H_{dS}^{d+1}} \frac{\sqrt{\pi}\Gamma\left(\frac{1+d}{2}\right)}{\Gamma(1+\frac{d}{2})} \\ &= -\frac{dV_{d+1}}{\kappa H_{dS}^{d-1}}. \end{aligned} \quad (1.24)$$

This should be seen as the semi-classical contribution to the dS free energy.

- If $\Lambda_{\text{eff}} < 0$: We have an asymptotically EAdS space and thus the action should diverge when approaching the boundary at $\tau \rightarrow \infty$. We thus regulate the action by fixing the radius of the boundary $a(\tau_f) = R$

$$\begin{aligned} S_{\text{bulk}}^{\text{on-shell}} &= -\frac{2V_d\Lambda_{\text{eff}}}{\kappa(D-2)} \int d\tau a^d = -\frac{2V_d\Lambda_{\text{eff}}}{\kappa(D-2)H_{AdS}^{d+1}} \int_0^{H_{AdS}R} du \frac{u^d}{\sqrt{1+u^2}}. \\ &= -\frac{2V_d\Lambda_{\text{eff}}}{\kappa(D-2)} \frac{R^{d+1}}{d+1} {}_2F_1\left(\frac{1}{2}, \frac{1+d}{2}, \frac{3+d}{2}, -(H_{AdS}R)^2\right). \end{aligned} \quad (1.25)$$

The Gibbons-Hawking yields

$$S_{GHY} = -\frac{dV_d\sqrt{1+H_{AdS}^2R^2}R^{d-1}}{\kappa}. \quad (1.26)$$

Combining these two contributions yield

$$S^{\text{on-shell}} = -\frac{2V_d\Lambda_{\text{eff}}}{\kappa(D-2)} \frac{R^{d+1}}{d+1} {}_2F_1\left(\frac{1}{2}, \frac{1+d}{2}, \frac{3+d}{2}, -(H_{AdS}R)^2\right) - \frac{dV_d\sqrt{1+H_{AdS}^2R^2}R^{d-1}}{\kappa}. \quad (1.27)$$

- If $\Lambda_{\text{eff}} = 0$: the bulk on-shell action trivially vanishes. We still need to regularize the integral because the GHY term diverges. We thus obtain

$$S^{\text{on-shell}} = -\frac{dV_dR^{d-1}}{\kappa}. \quad (1.28)$$

Perturbative solutions We focus on the 1-instanton sector and thus assume the potential has the following form

$$V_{\text{np}}(\chi) = \lambda \cos(\chi f_\alpha), \quad (1.29)$$

with $\lambda \ll 1$. We look for perturbative solutions in λ of the form

$$\begin{aligned} \chi(\tau) &= \chi_{(0)}(\tau) + \lambda \chi_{(1)}(\tau) + \lambda^2 \chi_{(2)}(\tau) + \dots \\ \alpha(\tau) &= a_{(0)}(\tau) + \lambda a_{(1)}(\tau) + \lambda^2 a_{(2)}(\tau) + \dots \end{aligned} \quad (1.30)$$

We assume charged solution

$$\begin{aligned} Q &= \int_{S^d} \star H = \int_{S^d} (\star)^2 d\chi \\ &= Q_{(0)} + \lambda Q_{(1)} + \lambda^2 Q_{(2)} + \dots \\ &= \int_{S^d} (\star)^2 d\chi_{(0)} + \lambda \int_{S^d} (\star)^2 d\chi_{(1)} + \lambda^2 \int_{S^d} (\star)^2 d\chi_{(2)}. \end{aligned} \quad (1.31)$$

There are two observations. First, the trivial set of solutions discussed previously have vanishing axion charge. Second, these solutions have running charges a priori. This can be seen by integrating the first equation on the compact slices of (1.22) yielding

$$\partial_\tau Q + (-1)^d V_d a^d \partial_\chi V_{\text{np}} = 0. \quad (1.32)$$

As one can observe the non-perturbative potential has the effect of non-conservation of the axionic charge. We now wish to describe a perturbative way to obtain the solution. We still discuss $k=1$ in the $N=1$ gauge. At order λ^0 , we obtain

$$\begin{aligned} \partial_\tau \left(a_{(0)}^d \partial_\tau \chi_{(0)} \right) &= 0, \\ \frac{d(d-1)}{a_{(0)}^2} \left(a_{(0)}'^2 - 1 \right) + 2\kappa \left(\frac{(\partial_\tau \chi_{(0)})^2}{2} + \frac{\Lambda}{\kappa} \right) &= 0. \end{aligned} \quad (1.33)$$

As we can observe the charge is constant at order λ^0 so $Q_{(0)} = \text{const}$. These equations become

$$\begin{aligned} a_{(0)}^d \partial_\tau \chi_{(0)} &= (-1)^d Q, \\ a_{(0)}'^2 &= 1 - \frac{2\kappa}{(a_0)^2 d(d-1)} \left(\frac{Q^2}{(V_d)^2 a_{(0)}^{2d}} + \frac{\Lambda}{\kappa} \right) \\ &= 1 - \frac{1}{a_{(0)}^2 H_{dS}^2} - \frac{2\kappa}{d(d-1)} \frac{Q^2}{(V_d)^2 a_{(0)}^{2d+2}}. \end{aligned} \quad (1.34)$$

Once $a_{(0)}$ is known, $\chi_{(0)}$ is also known.

At order λ^1 , one obtains

$$\begin{aligned} \partial_\tau \left(a_{(0)}^d \partial_\tau \chi_{(1)} + a_{(1)}^d \partial_\tau \chi_{(0)} \right) - a_{(0)}^d \lambda f_\alpha \sin(\chi_{(0)}) &= 0, \\ \frac{d(d-1)}{a_{(0)}^2} \left(a_{(0)}' a_{(1)}' - \frac{a_{(1)} (a_{(0)}')^2}{a_{(0)}} - \frac{a_{(1)} k}{a_{(0)}} \right) + 2\kappa \left(\partial_\tau \chi_{(0)} \partial_\tau \chi_{(1)} + \frac{\cos(\chi_{(0)} f_\alpha)}{2} \right) &= 0. \end{aligned} \quad (1.35)$$

Now the question is whether or not we can solve analytically these set of perturbative equations.

Numerical implementation In parallel, we can numerically integrate the first two equations of (1.22) which we recall here

$$\begin{aligned} \partial_\tau (a^d \partial_\tau \chi) - \lambda f_\alpha a^d \sin(\chi f_\alpha) &= 0, \\ \frac{d(d-1)}{a^2} (a'^2 - k) + 2\kappa \left(\frac{(\partial_\tau \chi)^2}{2} + \lambda \cos(\chi f_\alpha) + \frac{\Lambda}{\kappa} \right) &= 0. \end{aligned} \tag{1.36}$$

We want to solve for (χ, a) . Note that these equations are unchanged under a time reversal transformation. It should be sufficient to give as initial conditions the value of χ which we set at a local maximum of the potential at $\chi = 0$ WLOG. We also give an initial value for the scale factor $a(\tau = 0) = \mathbf{a} > \mathbf{0}$ as well as a velocity of the $\chi'(\tau = 0) = \chi'_0$ particle. **PG: You cannot consistently impose $\chi'(0) = 0 = a'(0)$ to obtain non trivial $\mathbb{Z}_2$ even solutions. Indeed, the first equation imposes $\sin(\chi f_\alpha) = 0$ at the initial time so χ lies at an extremum. Then, the second equation gives the value of the scale factor wrt Λ_{eff} .**